Course Code: CSE-1213 | Course Title: Discrete Mathematics

Full Marks: 70 | Time: 03 Hours

N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer **Six** questions, taking **Three** from each section **including Q.1, Q.2, Q.5 and Q.6**
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

Q.1 (a) Explain discrete and continuous objects and their differences with example. Describe the reasons of why should you study discrete mathematics? **[CO1, PO1, C2, KP1]** 3

(b) Explain Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$? **[CO2, PO2, C2, KP2]** 4

(c) Show that each of these conditional statements is a tautology by using truth tables. i) $(p \wedge q) \rightarrow (p \rightarrow q)$ ii) $\neg p \rightarrow (p \rightarrow q)$ **[CO3, PO3, C2, KP2]** 3

(d) Construct the prefix form for $((x + y) \uparrow 2) + ((x - 4)/3)$. **[CO4, PO3, C3, KP3]** 2

Q.2 (a) Let p and q be the propositions. Construct a truth table for each of these compound propositions $(p \wedge q) \rightarrow (p \vee q)$ **[CO3, PO3, C3, KP2]** 3

(b) Define De Morgan Laws. Compute that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent. **[CO3, PO3, C3, KP2]** 4

(c) Construct the truth value of $\forall x P(x)$, where P(x) is the statement "$x^2 < 10$" and the domain consists of the positive integers not exceeding 4? **[CO4, PO3, C3, KP3]** 5

Q.3 (a) Show that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using membership table. **[CO2, PO2, C2, KP2]** 3

(b) If A= {a, e, i, o, u}, C= {p, q, r, s, t, u, v, w, x, y, z} and X= {a, b, c,…., x, y, z} is the universal set, then calculate $(A \cup C)'$. **[CO2, PO2, C3, KP3]** 4

(c) Identify whether the relation R defined in the set {1, 2, 3, 4, 5, 6} as R = {(a,b): b = a + 1} is reflexive, symmetric or transitive. **[CO2, PO2, C3, KP3]** 4

Q.4 (a) Define proposition. Which of these sentences are propositions and why? 3

i) 5 + 7 = 10

ii) x + 2 = 11

iii) What time is it?

[CO1, PO2, C2, KP3]

(b) In a class of 50 students, 20 students play football, 16 students play hockey. It is found that 10 students play both the games. Calculate the number of students who play neither. Also construct a Venn Diagram to represent the information. **[CO2, PO2, C3, KP2]** 4

(c) Find out the ordered pairs in the less than or equal to relation, which contains (a, b) if $a \leq b$, on the set {0, 1, 2, 3}? **[CO2, PO2, C3, KP2]** 4

Section-B

Q.5 (a) Find the bitwise OR, AND, and XOR of the following bit strings:' A= 011011 0110 and B= 1100011101. **[CO2, PO2, C2, KP2]** 4

(b) Define Truth Set. What are the truth sets of the predicates P(x), Q(x), and R(x), where the domain is the set of integers and P(x) is "$|x| = 1$", Q(x) is "$x^2 = 2$" and R(x) is "$|x| = x$". **[CO2, PO2, C3, KP2]** 4

(c) Define direct and indirect proof. Let f_1 and f_2 be functions from R to R such that $f_1(x) = x_2$ and $f_2(x) = x - x^2$. What are functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f? **[CO4, PO4, C3, KP2]** 4

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Q.6 (a) Explain the degree of vertex for an undirected graph. How many edges are there in an undirected graph with 10 vertices of degree six? **[CO3, PO3, C2, KP3]** 3

(b) Define simple graph, multi graph and pseudograph. Draw the following graph: 5

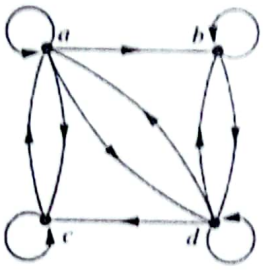
i) K_5 complete graph

ii) Bipartite graph $K_{3,6}$.

[CO4, PO4, C2, KP3]

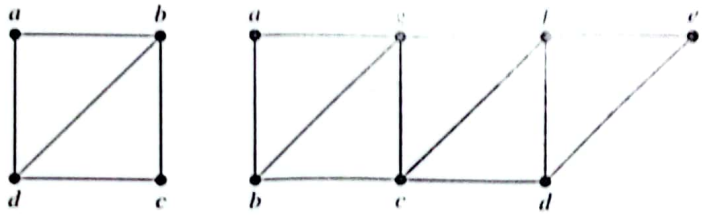
(c) Consider the following graph and determine the adjacency matrix of this graph. 4

[CO2, PO2, C3, KP3]



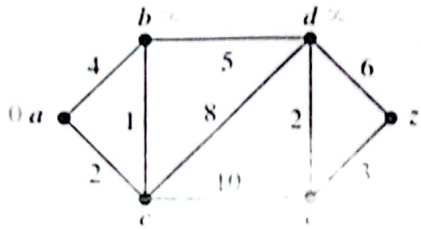
Q.7 (a) Define Euler path, Euler circuit. Identify whether the given graphs G_1 and G_2 have Euler path and circuit. If yes, write down at least one path and a circuit. 3

[CO3, PO3, C3, KP3]

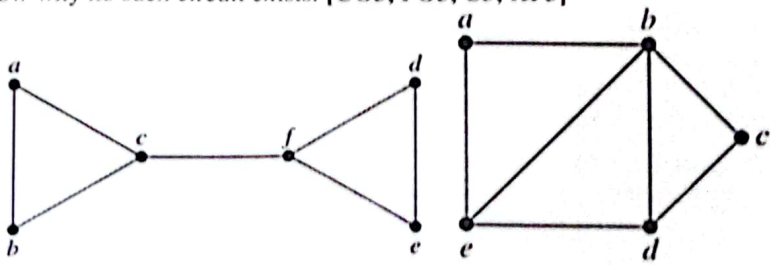


G_1 G_2

(b) Define weighted graph. Identify the shortest path from source vertex **a**, to all other vertex using Dijkstra's algorithm. **[CO3, PO3, C3, KP3]** 4



(c) Define Hamilton graph and circuit. Identify whether the given graphs have a Hamilton circuit. If it does, find such a circuit. If it does not, give an argument to show why no such circuit exists. **[CO3, PO3, C3, KP3]** 4



G_1 G_2

Q.8 (a) Consider the following binary tree, and explain both the ways to delete node 60 having two children and redraw the new tree. **[CO3, PO2, C2, KP3]** 3

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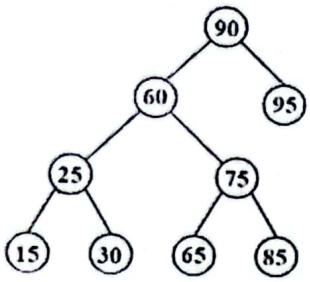
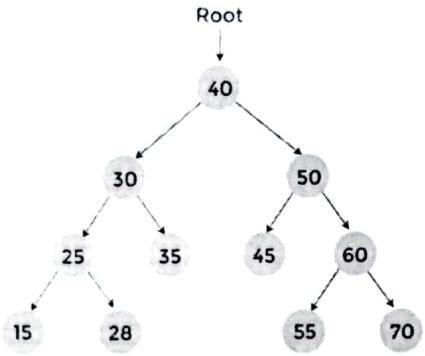


(b) Explain Binary search Tree. Find In-order and Post order traversal output of the following binary tree. **[CO3, PO2, C2, KP3]** **4**



(c) Define Polish notation. Construct the binary tree for the mathematical expression: $((x + y) \uparrow 2) + ((x - 4)/3)$. **[CO3, PO3, C3, KP3]** **4**

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Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering

B.Sc. Engg. 1st Year 2nd Semester, Referred/ Improvement/ Backlog Examination, Fall-2023

Course Code: CSE-1213 | **Course Title: Structured Programming**

Full Marks: 70 | **Time: 03 Hours**

N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer **Six** questions, taking **Three** from each section **including Q.1, Q.2, Q.5 and Q.6**
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

Q.1 (a) Demonstrate different types of variables with suitable examples. What happens when a floating-point number is kept in an integer space. **[CO2, PO2, C2, KP3]** 3

(b) Raya wants to develop a system that prints a message "Beautiful" (without quotes) if an integer number is divisible by 5 and 10. Otherwise, she wants to display "Not beautiful" (without quotes). Now develop a C program to implement the system. **[CO2, PO2, C3, KP4]** 4

(c) You are given the lengths of three sides of a triangle. Develop a program to determine the area of the triangle. **[CO2, PO2, C3, KP4]** 5

Q.2 (a) Construct a C program to find the output of the following series: 3

$$5^2 + 8^4 + 11^6 + 14^8 + \ldots\ldots\ldots\ldots\ldots\ldots + m^n$$

Here ***m*** and ***n*** represent positive integer number taken from user. **[CO2, PO2, C3, KP3]**

(b) Raya learnt that a strong number is one in which the factorial of each digit of a number equals the original number's sum. For example, 145 is a strong number because the sum of its factorials is equal to the number itself 1! + 4! + 5! = 145. Now she seeks your help to build a code that will determine whether any given input ***n*** is a strong number or not. **[CO2, PO2, C3, KP4]** 5

(c) Analyze the output for the following code snippet: **[CO2, PO2, C4, KP4]** 4

```
#include<stdio.h>
          int main(){
   for(int i=1; i<=5; i++){
      for(int j=1; j<=5-i; j++){
         printf("*");
      }
      printf("\n");
   }
}
```

Q.3 (a) Given an array of size ***n***. Now take ***n*** number of elements from user and construct a C program to calculate the summation of all the elements of the array. (Size and elements of the array will be taken from user). **[CO2, PO2, C3, KP4]** 3

(b) Suppose, there is an array M whose elements are only integers. Now develop a program to find the maximum number from the elements of the array. **[CO2, PO2, C3, KP4]** 4

(c) Analyze and then construct a code to determine the minimum number of swaps required in case of **Bubble Sort** to sort an array ***A*** in ascending or descending order of size ***n***. Size and elements of the array will be taken from user. **[CO2, PO2, C4, KP4]** 4

Q.4 (a) Explain how dereferencing becomes useful in C programming. **[CO3, PO3, C2, KP3]** 3

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(b) Analyze and find the output of the following code snippet: **[CO3, PO3, C4, KP5]** 3

```
#include<stdio.h>
int main(){
  int v = 385;
  int *p;
  p = &v;
  char *p1;
  p1 = p;
  printf("%d", *(p1+1));
}
```

(c) "In case of passing an array to a function, the whole array doesn't get passed to that function rather only the reference gets passed" – do you agree with this statement? Give proper example to justify your answer. **[CO3, PO3, C5, KP5]** 5

Section-B

Q.5 (a) Illustrate the output of the following program: **[CO2, PO2, C2, KP4]** 3

i.

```
#include<stdio.h>
int main ()
{
  int a=1, b=-1, c=0, d;
  d=++a && ++b || c-- ;
  if (d)
        printf("Kolkata \n");
  else if (c)
        printf("Delhi \n");
  else
       printf("Bangalore \n");
  return 0;
}
```

ii.

```
#include<stdio.h>
int main()
{
     struct simp
     {
          int i = 6;
          char city[] = "chennai";
     };
    struct simp s1;
    printf("%d",s1.city);
    printf("%d", s1.i);
    return 0;
}
```

(b) In the small town of Rainville, the inhabitants rely heavily on rainfall for their water supply and agricultural needs. Each year, the town's weather department meticulously records the total rainfall for each month to monitor and analyze patterns. This data helps farmers plan their crops, the town council to manage water resources, and the residents to prepare for dry or wet months. 4

You are a programmer tasked with helping the Rainville Weather Department by writing a program that processes this rainfall data. Now, construct a that will take as input the total rainfall for each of the twelve months and then calculate and display the total rainfall for the year, the average monthly rainfall, and the months with the highest and lowest amounts of rainfall.

Sample Input: 75 85 90 120 110 95 100 80 65 70 85 95
Sample Output:
Total rainfall for the year: 1070
Average monthly rainfall: 89.17
Month with the highest rainfall: April (120)
Month with the lowest rainfall: September (65)
[CO2, PO2, C3, KP4]

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3 (c) In the ancient kingdom of Palindoria, there exists a magical library filled with scrolls containing powerful spells and ancient knowledge. According to legend, only those who possess the wisdom to recognize a special type of word, known as a palindrome, can unlock the secrets of the library. **5**

The Guardian of the Library, a wise and powerful entity, has set a challenge for those who seek entry. The Guardian presents you with a string and demands that you determine whether it is a palindrome. If you can correctly identify the nature of the string, you will be granted access to the library's secrets.

Your task is to construct a program that takes a string as input and returns "Yes, it is" if the given string is a palindrome or "No, it isn't" if it is not a palindrome.

[CO3, PO3, C3, KP5]

Sample Input 1: A man a plan a canal Panama **Sample Output 1:** Yes, it is	**Sample Input 2:** Hello **Sample Output 2:** No, it isn't

Q.6 (a) Outline the errors that occurred in the program below and translate it to the acceptable version. **[CO2, PO2, C2, KP3]** **3**

```
#include <studio.h>
int main()
    int number1, number2
    printf("Enter two integers: ");
    scanf("%d d", &number1, number2);
    if(number1 == number2){
        printf("%d %d",&number1,number2);
    }
    else if (number1 > number2){
        print("%d", number1, number2);
    }
    else {
        printf(" d  %d",number1, &number2)
    }
    return 0;
```

(b) One day, a young and talented programmer named Elara embarks on a quest to find the Hidden Rune. After solving numerous riddles and navigating through labyrinthine corridors, Elara finally arrives at the entrance of the Codex Chamber. There, she is met by an ancient guardian spirit who presents her with the final challenge: **4**

"To enter the Codex Chamber and claim the Hidden Rune, you must prove your mastery of strings. I will give you a magical input device. You must write a function that processes the strings I provide. Here are the rules:

i. You will receive a string from me.
ii. If the length of the string is exactly 7 characters, you must display the middle character of that string.
iii. If the string is not 7 characters long, you must prompt for another string until you get a string of the correct length.

Now construct the program based on the given rules using function.

[CO2, PO2, C3, KP4]

(c) In the bustling city of Codeburg, there lived a renowned detective named Sherlock Coder. One day, Sherlock received an urgent message from the mayor of Codeburg, stating that all the uppercase letters and digits in the city had suddenly vanished! Panic spread through the streets as citizens struggled to communicate without their beloved uppercase letters and digits. **5**

Determined to solve the case, Sherlock embarked on a quest to restore the missing letters. Armed with his trusty programming skills, he began his investigation by examining a series of strings from different locations in the city.

Your task is to assist Sherlock Coder by developing a program to remove all uppercase letters and digits letters from a given string. The input will consist of a single string containing a mixture of lowercase and uppercase letters, along with other characters. Your program should remove all uppercase letters and digits from the string and output the result. **[CO3, PO3, C3, KP5]**

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Sample Input: CodeburgIsInCrisis123!
Sample Output: odeburgsnrisis!

Q.7 (a) Explain comment and its types with proper example. **3**
[CO2, PO2, C2, KP3]

(b) In the bustling city of Codeopolis, there's an annual programming competition known as the "Digit Detective Challenge." This year, the organizers have come up with an intriguing task for the participants. **4**
You are given the task of creating a program to assist the Codeopolis Digital Security Agency (CDSA). The CDSA needs a reliable way to verify certain identification numbers entered by agents. These identification numbers must be exactly four digits long to be considered valid. If a number is valid, the system should grant access with a welcoming message. Otherwise, it should deny access with a rejection message.
Construct a program that takes a single integer as input and checks if it is a four-digit number. If it is, your program should display the message "Welcome." If it is not a four-digit number, your program should display the message "rejected."
[CO2, PO2, C3, KP4]

Sample Input 1: 1234 **Sample Output 1:** Welcome	**Sample Input 2:** 999 **Sample Output 2:** rejected

(c) Floyd's Triangle is a triangular array of natural numbers arranged in a specific order. Your task is to construct a program that takes a number n as input and prints the first n rows of Floyd's Triangle. **[CO2, PO2, C3, KP5]** **4**

Sample Input: 5
Sample Output:
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

Q.8 (a) Illustrate the advantages and disadvantages of using structure over array. **3**
[CO3, PO3, C2, KP5]

(b) You are an IT technician in a health clinic. You have been asked to write a sample C program that would help the doctor capture some data about a patient. Ensure your program captures data pertaining to: patient's name, age, eye colour, weight, skin colour and hair colour. Now your task is to construct a program using Union that will take patient's information as input and display it. **4**
[CO3, PO3, C3, KP5]

(c) Distinguish between structure and Union. **[CO3, PO3, C4, KP4]** **4**

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Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering

B.Sc. Engg. 1st Year 2nd Semester Referred/Improvement/Backlog Examination Fall-2023

Course Code: EEE 1261 | **Course Title: Electronic Devices and Circuits**

Full Marks: 70 | **Time: 03 Hours**

N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer **Six** questions, taking **Three** from each section **including Q.1, Q.2, Q.5 and Q.6**
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

			Marks
Q.1	(a)	Define the terms: **[CO1, PO1, C1, K3]** (i) Depletion Region, (ii) Peak Inverse Voltage.	3
	(b)	Explain the forward and reverse bias in a PN junction. **[CO2, PO1, C2, K3]**	4
	(c)	Calculate the current through 48 Ω resistor in the circuit shown in Fig. 1 (c). Assume the diodes to be of silicon and forward resistance of each diode is 1.5 Ω. **[CO3, PO3, C3, K5]**	5

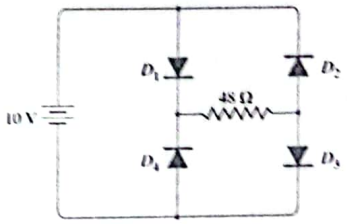

Fig. 1 (c)

Q.2	(a)	Briefly explain the roles of majority and minority charge carriers in n-type and p-type semiconductors. **[CO2, PO1, C2, K3]**	3
	(b)	Illustrate the V-I characteristic curve of a p-n junction diode and label the various regions. **[CO1, PO1, C2, K3]**	4
	(c)	Determine V_0 for the network of Fig. 2 (c). **[CO3, PO3, C3, K5]**	5

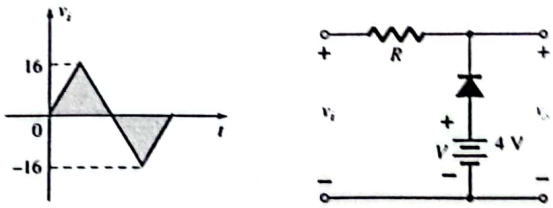

Fig. 2 (c)

Q.3	(a)	Describe the purpose and operation of a clipping circuit and provide an example of its use.	3
	(b)	Compare and contrast half-wave and full-wave rectifiers in terms of efficiency and ripple factor.	4
	(c)	Draw the output waveshape for the following circuit of Fig. 3 (c). Where, $V = 10V, C = 1\mu F, R = 100K\Omega$ and $V_1 = 5V$.	4

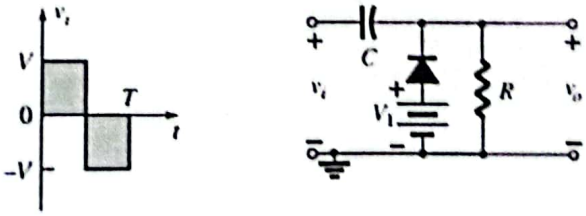

Fig. 3 (c)

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Q.4 (a) Define an operational amplifier and list its applications. 3

(b) Explain the characteristics of an ideal Op-Amp. Why do we use feedback resistor in an op-amp. 3

(c) Design an operational amplifier circuit for the following output voltage: 5

$$V(0) = 10\,V(1) - 4\,V(2) + 6\,V(3) - 8\,V(4)$$

Section-B

Q.5 (a) Describe the basic construction of a bipolar junction transistor (BJT). **[CO1, PO1, C1, K3]** 3

(b) Deduce the relationship between α and β in a BJT, specifying the significance of each symbol. **[CO2, PO1, C2, K3]** 4

(c) Design a logic gate by using the transistor that produces the waveform shown in Fig. 5 (c). **[CO3, PO3, C6, K5]** 5

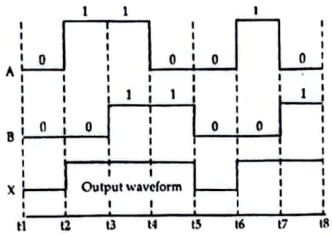

Fig. 5 (c)

Q.6 (a) Explain the operation of CMOS as an inverter. **[CO2, PO1, C2, K3]** 4

(b) What is the significance of DC biasing in a transistor? Illustrate the input and output characteristic curves of a common-base BJT. **[CO1, PO1, C2, K3]** 3

(c) For the network of Fig. 6 (c). **[CO3, PO3, C3, K3]** 5

a. Find I_B, I_C, and I_E.
b. Determine V_B, V_C, and V_E.
c. Calculate V_{BC} and V_{CE}

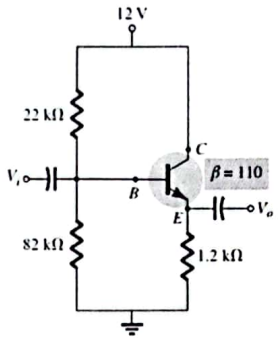

Fig. 6 (c)

Q.7 (a) Define "FET". Write the applications of FETs. 3

(b) Explain the difference between BJT and FET. 4

(c) Determine the following for the network of Fig. 7 (c). 4

a. V_{GSQ}
b. I_{DQ}
c. V_{DS}
d. V_D
e. V_G
f. V_S

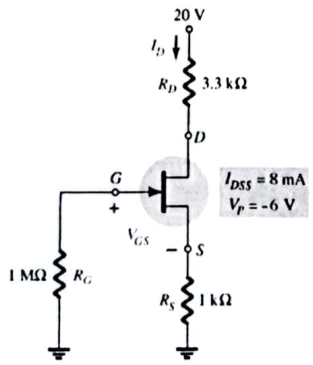

Fig. 7 (c).

Q.8 (a) Define the terms: DIAC, SCR and UJT. 3

(b) Briefly explain the operation of a TRIAC with a circuit diagram and input-output waveform. 4

(c) Outline the basic steps involved in the fabrication of an integrated circuit. 4

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Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering

B.Sc. Engg. 1st Year 2nd Semester Referred/Improvement/Backlog Examination Fall-2023

Course Code: EEE-1261 | **Course Title: Electrical Circuit Analysis**

Full Marks: 70 | **Time: 03 Hours**

N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer **Six** questions, taking **Three** from each section **including Q.1, Q.2, Q.5 and Q.6**
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

Q.1 (a) State one Kirchhoff's Law that is used in Mesh analysis and also explain this Law with proper diagram. **[CO1, PO1, C1, K2]** 3

(b) Explain the CDR and KVL with proper example. **[CO1, PO1, C2, K2]** 4

(c) Determine I_1 & I_2 in the circuit of Fig. 1(c) and count the number of node, mesh and loop. **[CO1, PO1, C3, K3]** 5

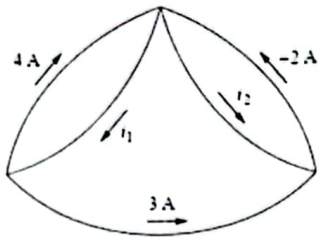

Fig. 1(c)

Q.2 (a) Briefly discuss the voltage divider rule with circuit diagram. **[CO1, PO1, C2, K2]** 3

(b) Explain how does capacitor and inductor behaves on DC supply with proper example. **[CO3, PO3, C3, K4]** 3

(c) Find currents and voltages for Fig. 2(c). **[CO1, PO1, C3, K3]** 6

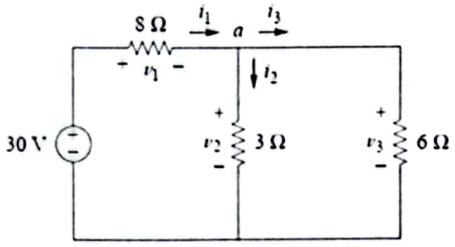

Fig. 2(c)

Q.3 (a) Explain why node and mesh analysis may not be suitable for all electrical circuits and also approaches to overcome this limitations. 3

(b) Determine the node voltages in the following circuit in Fig. 3(b) 4

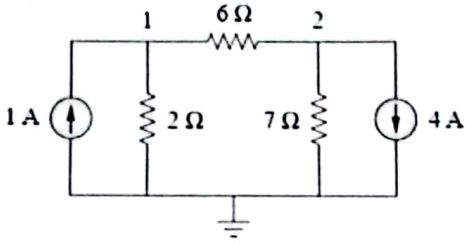

Fig. 3(b)

(c) Use the superposition theorem to find v in the circuit in Fig. 3(c). 4

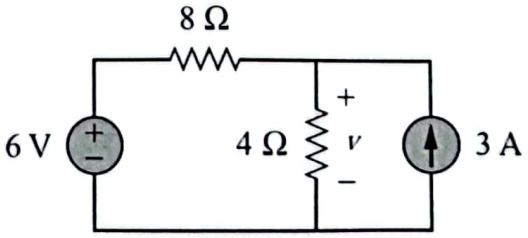

Fig. 3(c)

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Q.4 (a) State and explain Superposition theorem. 3

(b) State and explain Thevenin's theorem. Why do we need this theorem? Draw its equivalent circuit. 4

(c) Determine the Thevenin equivalent circuit diagram of the circuit in Fig. 4 (c). 4

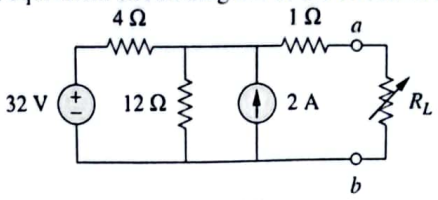

Fig. 4 (c)

Section-B

Q.5 (a) State maximum power transfer and maximum average power transfer theorem with example. **[CO2, PO2, C1, K2]** 3

(b) Derive the condition of maximum power transfer theorem and from this also prove that **[CO2, PO2, C2, K3]** 4

$$P_{max}=V_{th}^2/(4\,R_{th})$$

(c) Find the value of R_L for maximum power transfer in the circuit given in Fig. 5(c) and also determine that power. **[CO2, PO2, C3, K3]** 5

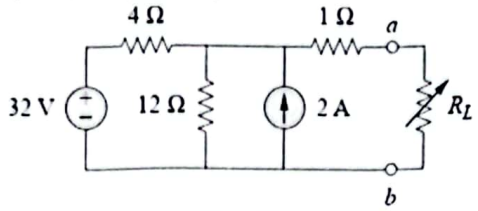

Fig. 5(c)

Q.6 (a) Draw impedance and power triangle with proper specification. **[CO2, PO2, C2, K2]** 3

(b) Find the ***phase angle*** between $V_1 = -4\sin(37t + 25^0)$ and $V_2 = -5\cos(37t + 40^0)$. And also determine which one is lead? **[CO2, PO2, C3, K3]** 4

(c) In Fig. 6(c) find Z_{in}, Y_{in} & B_{in} and also find the voltage $V_0(t)$. **[CO2, PO2, C3, K3]** 5

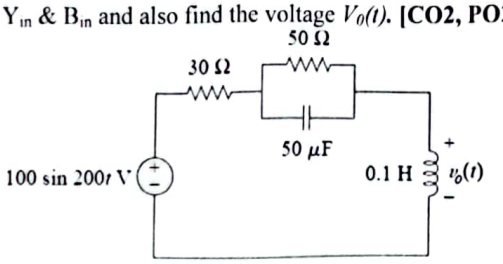

Fig. 6(c)

Q.7 (a) Write down the voltage and current relationship of resistor, inductor and capacitor. 3

(b) Calculate the total inductance L_{AB} in the following circuit in Fig. 7(b). 4

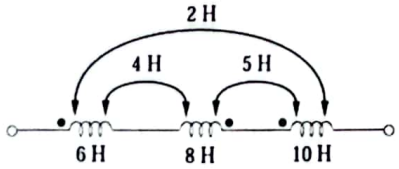

Fig. 7(b)

(c) Derive the power equations for both Δ (delta) and Y (wye) connected three-phase circuits. 4

Q.8 (a) Discuss the advantages of 3φ system over 1φ system. 3

(b) Show that the 3φ system uses only 75% of the material those used in the equivalent 1φ system. 4

(c) Find the expression of generalized line currents of the Balanced Y-Y networks. 4

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Bangladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering

B.Sc. Engg. 1st Year 2nd Semester, Referred/Improvement/Backlog Examination, Fall-2023

Course Code: MATH-1241

Course Title: Ordinary, Partial Differential Equations and Coordinate Geometry

Full Marks: 70 **Time: 03 Hours**

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer **Six** questions, taking **Three** from each section **including Q.1, Q.2, Q.5 and Q.6.**
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

Marks

Q.1 **(a)** What do you understand by order and degree of a differential equation? Find the order and degree of the ordinary differential equation $\left(\frac{d^3y}{dx^3}\right)^2 + 5\frac{d^3y}{dx^3} = 6\frac{dy}{dx} - y$. **[CO1, PO1, C2, KP1]** **4**

(b) Form an ordinary differential equation of all circles passing through the origin and having center on the x-axis. **[CO1, PO1, C5, KP2]** **4**

(c) The rate at which ice melts is proportional to the amount of ice at that instant. Find the temperature of ice left after 2 hours if half of the ice melts in 30 minutes. **[CO2, PO2, C5, KP3]** **4**

Q.2 Compute the general solution of any three of the following differential equations, including i: **[CO2, PO2, C5, KP3]** **4+4+4**

i. $(e^y + 1)\cos x\, dx + e^y \sin x\, dy = 0$

ii. $x\cos(\frac{y}{x})(ydx + xdy) = y\sin(\frac{y}{x})(xdy - ydx)$

iii. $(x + 2y^3)\frac{dy}{dx} = y$

iv. $x\frac{dy}{dx} + y = y^2 \log x$

Q.3 Determine the complementary function, particular integral and hence obtain the complete solution of the following differential equations (any three including i): **3+4+4**

i. $(D^3 - D^2 - 4D + 4)y = 0$

ii. $(D^2 - 3D + 2)y = e^x + x^2$

iii. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = \cos 2x$

iv. $(D^2 - 4D + 3)y = x^2 \sin 3x$

Q.4 **(a)** Explain the partial differential equation. Construct a partial differential equation of the set of all right circular cones whose axes coincide with z axis. **5**

(b) Convert the following relationship $\phi\left(2x - 3y - z^2,\ x^2 + y\right) = 0$ to partial differential equation by eliminating the arbitrary function ϕ. What is the order of the obtained partial differential equation? **6**

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Section-B

			Marks
Q.5	(a)	Note down the procedure involved in applying Lagrange's method to the partial differential equation of $Pp+Qq=R$, where P, Q, and R are functions of x, y, and z. **[CO3, PO2, C2, KP1]**	4
	(b)	Solve the following partial differential equations: **[CO3, PO2, C5, KP4]**	4+4
		i. $\frac{y^2 z}{x}p+xzq=y^2$	
		ii. $(mz-ny)p+(nx-lz)q=ly-mx$	
Q.6	(a)	Evaluate the integral surface of the linear partial differential equations $x\left(y^2+z\right)p-y\left(x^2+z\right)q=\left(x^2-y^2\right)z$ which contains the straight line $x+y=0$ and $z=1$. **[CO3, PO2, C5, KP4]**	6
	(b)	Find the complete integral of the partial differential equation $xp+yq=pq$. **[CO3, PO2, C5, KP4]**	6
Q.7		Identify the general solution of any three of the following partial differential equations, including i:	3+4+4
		i. $(D^3D'^2+D^2D'^3)z=0$	
		ii. $(D^2D'+DD'^2)z=3\sqrt{2x-3y}$	
		iii. $(D^2-DD'-6D'^2)z=\cos(x+6y)$	
		iv. $(D^2-7DD'-3D'^2)z=e^{2y}\sin 3x$	
Q.8	(a)	When does the general equation of second degree become circle, ellipse, parabola, and hyperbola?	3
	(b)	The equation $3x^2+2xy+3y^2-18x-22y+50=0$ is transformed to $4x^2+2y^2=1$ when referred to rectangular axes through the point $(2,3)$. Find the inclination of the later axes to the former.	4
	(c)	Execute the equation of the plane passing through the intersection of the planes $x+2y+3z+4=0$ and $4x+3y+2z+1=0$ and the point $(1,2,3)$.	4

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